

Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H\psi(x, t)$$

Let $\psi(x, t)$ be separable as

$$\psi(x, t) = \phi(t)\varphi(x)$$

Let it be the case that for the time derivative we have

$$i\hbar \frac{\partial}{\partial t} \phi(t) = E\phi(t)$$

where E is a constant. Then

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = E\phi(t)\varphi(x) = E\psi(x, t)$$

Hence $\psi(x, t)$ is a solution to the time-independent Schrodinger equation

$$E\psi(x, t) = H\psi(x, t)$$

If H is independent of t then we also have

$$E\varphi(x) = H\varphi(x)$$

For example, the Hamiltonian for the harmonic oscillator is

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2$$

The wavefunctions are

$$\psi_n(x, t) = C_n \exp\left(-\frac{iE_n t}{\hbar}\right) \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is a normalization constant and

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

Taking the time derivative of $\psi_n(x, t)$ we find

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = E_n \psi(x, t)$$

Hence $\psi(x, t)$ is a solution to the time-independent Schrodinger equation

$$E_n \psi(x, t) = H\psi(x, t)$$

Eigenmath script